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Spinous process fixation system and methods thereof

Abstract

An implantable device may comprise a barrel, the barrel having an upper portion and a lower portion. The barrel may be configured to transition from a collapsed form having a first height to an expanded form having a second height and wherein the second height is greater than the first height. The implantable device may further include an actuator assembly disposed in the barrel, the actuator assembly comprising a front ramped actuator in engagement with the barrel, a rear ramped actuator in engagement with the barrel, and a central screw that extends from the rear ramped actuator through the front ramped actuator. The implantable device may further comprise a first plate and a second plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/379,619, filed Jul. 19, 2021, which is a continuation of U.S. patent application Ser. No. 16/258,017, filed Jan. 25, 2019, which is a continuation of U.S. patent application Ser. No. 15/268,995, filed on Sep. 19, 2016 (published as U.S. Pat. Pub. No. 2017/0000529), which is a divisional of U.S. patent application Ser. No. 14/057,946, filed Oct. 18, 2013 (now U.S. Pat. No. 9,486,251), which is a continuation-in-part of U.S. patent application Ser. No. 13/799,364, filed Mar. 13, 2013 (now U.S. Pat. No. 9,198,697), which is a continuation-in-part of U.S. patent application Ser. No. 13/731,504, filed Dec. 31, 2012 (now U.S. Pat. No. 9,011,493), the entire disclosures of which are incorporated herein by reference for all purposes.

FIELD OF THE INVENTION

(1) This description relates to medical devices and systems and more particularly to a spinous process fixation system and methods thereof. In particular, in one or more implementations, this description relates to spinous process fusion devices that distract and/or immobilize the spinous processes of adjacent vertebrae.

BACKGROUND

(2) A variety of medical devices and medical device systems may be implanted within a body of a patient to provide support to a portion or portions of the patient's body. For example, some medical devices may be implanted and coupled to backbones or portions of a spine of a patient and may be configured to provide support to the spinal bone structure of the patient.

(3) Typically, weaknesses in the spine are corrected using devices that fuse one or more vertebrae together. It may be desirable to have an implantable device that provides for structural stability to adjacent vertebrae and to achieve supplemental fusion to treat weaknesses in the spine due to degenerative disc disease, spondylolisthesis, trauma (i.e., fracture or dislocation), tumor and/or other causes.

SUMMARY

(4) According to one general aspect, an implantable device includes a barrel. The barrel has a first portion and a second portion. The implantable device includes a first plate having multiple projections extending from one side of the first plate, where the first plate is configured to movably couple to the first portion of the barrel. The implantable device includes a second plate having multiple projections extending from one side of the second plate, where the second plate is configured to movably couple to the second portion of the barrel. The barrel is configured to transition from a collapsed form having a first height to an expanded form having a second height, where the second height is greater than the first height.

(5) Implementations may include one or more of the following features. For example, the barrel may include a frame, a first endplate having a curved shape and a second endplate having a curved shape. The first endplate and the second endplate may be coupled to the frame to form the barrel, where the barrel has a bulleted shape in both a lateral direction and a posterior direction. The barrel may include a frame, a first endplate, a second endplate, a first actuator having a split ramp inserted into the frame, a second actuator having a split ramp inserted into the frame and a central screw inserted through the first actuator and the second actuator, where the first actuator and the second actuator are configured to act on the first endplate and the second endplate in response to a rotation of the central screw. The barrel may include a first window and a second window, where the first window and the second window may be configured to receive graft packing material. The barrel may include a first endplate having a shaped groove and a second endplate having a shaped groove.

(6) For example, in one implementation, the first portion and the second portion may be rails that extend from a same side of the barrel. For example, in another implementation, the first portion and the second portion may be rails that each extend from a different side of the barrel.

(7) For example, the first plate and the second plate are each shaped in a lordotic profile. The first plate may include a bushing to enable the first plate to angulate about the bushing and the second plate may include a bushing to enable the second plate to angulate about the bushing. The first plate may be locked in position using a first set screw at any position within a range of motion for the first plate and the second plate may be locked in position using a second set screw at any position within a range of motion for the second plate. The first set screw may include a cup-shaped end to lock the first plate in position and the second set screw may include a cup-shaped end to lock the second plate in position.

(8) In another general aspect, an implantable device includes a barrel having a first portion and a second portion, a first plate having multiple projections extending from one side of the first plate, where the first plate is configured to movably couple to the first portion of the barrel and to

angulate about an axis of the first portion, and a second plate having multiple projections extending from one side of the second plate, where the second plate is configured to movably couple to the second portion of the barrel and to angulate about an axis of the second portion. The first plate and the second plate are each shaped in a lordotic profile.

(9) Implementations may include one or more of the following features. For example, the first plate may be configured to angulate up to about 25 degrees about the axis of the first portion and the second plate may be configured to angulate up to about 25 degrees about the axis of the second portion. In one implementation, the first portion and the second portion may be rails that extend from a same side of the barrel. In another implementation, the first portion and the second portion may be rails that each extend from a different side of the barrel.

(10) For example, the barrel may be configured to transition from a collapsed form having a first height to an expanded form having a second height, where the second height is greater than the first height. The first plate may be locked in position using a first set screw at any position within a range of motion for the first plate, where the first set screw has a cup-shaped end, and the second plate may be locked in position using a second set screw at any position within a range of motion for the second plate, where the second set screw has a cup-shaped end.

(11) In another general aspect, a method includes inserting a barrel of an implantable device into an interspinous space. The implantable medical device includes the barrel having a first portion and a second portion, a first plate having multiple projections extending from one side of the first plate and a second plate having multiple projections extending from one side of the second plate. The method includes expanding the barrel from a collapsed form having a first height to an expanded form having a second height, where the second height is greater than the first height, moving the first plate on the first portion to engage a spinous process and moving the second plate on the second portion to engage the spinous process.

(12) Implementations may include one or more of the following features. For example, the method may include engaging set screws in the first plate and the second plate to lock the first plate and the second plate in position. The method may include positioning the first plate to a desired angle with respect to the first portion, positioning the second plate to a desired angle with respect to the second portion and engaging set screws in the first plate and the second plate to lock the first plate and the second plate in position.

(13) In another general aspect, an implantable device may include a barrel, the barrel having an upper portion and a lower portion. The implantable device may further include an actuator assembly disposed in the barrel, the actuator assembly comprising a front ramped actuator in engagement with the barrel, a rear ramped actuator in engagement with the barrel, and a central screw that extends from the rear ramped actuator through the front ramped actuator. The implantable device may further include a first plate having multiple projections extending from one side of the first plate, the first plate comprising a first portion that extends from the upper portion and a second portion that extends from the lower portion. The implantable device may further include a second plate having multiple projections extending from one side of the second plate, the second plate configured to be received on the central screw. The barrel may be configured to transition from a collapsed form having a first height to an expanded form having a second height and wherein the second height is greater than the first height.

(14) In another general aspect, a method may include implanting a medical device in a patient, the method comprising: inserting a barrel of the device between adjacent spinous process, the medical device comprising a first plate disposed on one end of the barrel; rotating a central screw disposed in the barrel to cause the barrel to expand from a collapsed form having a first height to an expanded form having a second height; and ratcheting a second plate onto the central screw such that the first plate and the second plate engage the adjacent spinous process, the second plate being free to rotate about its center within a range of motion.

(15) In another general aspect, an implantable device may comprise a barrel, the barrel having an

upper portion and a lower portion. The implantable device may further an actuator assembly disposed in the barrel, the actuator assembly comprising a front ramped actuator in engagement with the barrel, a rear ramped actuator in engagement with the barrel, and a central screw that extends from the rear ramped actuator through the front ramped actuator. The implantable device may further comprise a first plate. The first plate may comprise a first portion that extends from the upper portion, a second portion that extends from the lower portion, and a pivoting spike assembly disposed in the first plate. The pivoting spike assembly may comprise multiple projections extending from a first side of the first plate. The implantable device may further comprise a second plate having multiple projections extending from a first side of the second plate, the second plate configured to be received on the central screw. The barrel may be configured to transition from a collapsed form having a first height to an expanded form having a second height and wherein the second height is greater than the first height.

(16) In another general aspect, a method for implanting a medical device in a patient may comprise: inserting a barrel of the device between adjacent spinous processes, the medical device comprising a first plate disposed on one end of the barrel; rotating a central screw disposed in the barrel to cause the barrel to expand from a collapsed form having a first height to an expanded form having a second height; and inserting a second plate onto the central screw such that the first plate and the second plate engage the adjacent spinous process wherein at least one of the first plate or the second plate comprises a pivoting spike assembly that is allowed to freely articulate until engagement with the adjacent spinous process compresses the pivoting spike assembly locking the pivoting spike assembly in place.

(17) In another general aspect, a method of assembling a medical device may comprise providing a medical device comprising a first plate extending from at least one side of a central barrel; inserting a spike assembly into an opening in the first plate such that a hole in the spike assembly is aligned with a hole in the first plate; inserting a fastener through the hole in the spike assembly and the hole in the first plate such that a lobe of the fastener engages the spike assembly; rotating the fastener while disposed in the hole of the spike assembly and the hole in the first plate such that the fastener causes a leaf spring feature of the spike assembly to expand; and further inserting the fastener through the hole in the spike assembly and the hole in the first plate to push the lobe below the leaf spring feature and into a groove in the spike assembly.

(18) The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a medical device according to an example implementation.
- (2) FIG. 2 is a top view of the medical device of FIG. 1.
- (3) FIG. 3 is a front view of the medical device of FIG. 1.
- (4) FIG. 4 is a side view of the medical device of FIG. 1.
- (5) FIG. 5 is a perspective view of a medical device according to an example implementation.
- (6) FIG. 6 is a top view of the medical device of FIG. 5.
- (7) FIG. 7 is a front view of the medical device of FIG. 5.
- (8) FIG. 8 is a side view of the medical device of FIG. 5.
- (9) FIG. 9 is a perspective view of a medical device according to an example implementation.
- (10) FIG. 10 is a top view of the medical device of FIG. 9.
- (11) FIG. 11 is a front view of the medical device of FIG. 9.
- (12) FIG. 12 is a side view of the medical device of FIG. 9.

- (13) FIG. 13 is a perspective view of a medical device according to an example implementation.
- (14) FIG. 14 is a top view of the medical device of FIG. 13.
- (15) FIG. 15 is a front view of the medical device of FIG. 13.
- (16) FIG. 16 is a side view of the medical device of FIG. 13.
- (17) FIG. 17 is a perspective view of a medical device according to an example implementation.
- (18) FIG. 18 is a top view of the medical device of FIG. 17.
- (19) FIG. 19 is a front view of the medical device of FIG. 17.
- (20) FIG. 20 is a side view of the medical device of FIG. 17.
- (21) FIG. 21 is a top view of a medical device according to an example implementation.
- (22) FIG. 22 is a side view of the medical device of FIG. 21.
- (23) FIG. 23 is a detailed view of the inset A of FIG. 22.
- (24) FIGS. 24-27 are side views of a plate of a medical device according to an example implementation.
- (25) FIG. 28 is top view of a barrel of a medical device according to an example implementation.
- (26) FIG. 29 is an exploded front view of a barrel of a medical device according to an example implementation.
- (27) FIG. 30 is an exploded top view of a barrel of a medical device according to an example implementation.
- (28) FIG. 31 is a perspective view of a medical device according to an example implementation.
- (29) FIG. 32 is a top view of the medical device of FIG. 31.
- (30) FIG. 33 is a front view of the medical device of FIG. 31.
- (31) FIG. 34 is a side view of the medical device of FIG. 31.
- (32) FIG. 35 is a perspective view of a medical device according to an example implementation.
- (33) FIG. 36 is a top view of the medical device of FIG. 35.
- (34) FIG. 37 is a front view of the medical device of FIG. 35.
- (35) FIG. 38 is a side view of the medical device of FIG. 35.
- (36) FIG. 39 is a perspective view of a medical device according to an example implementation.
- (37) FIG. 40 is a top view of the medical device of FIG. 39.
- (38) FIG. 41 is a front view of the medical device of FIG. 39.
- (39) FIG. 42 is a side view of the medical device of FIG. 39.
- (40) FIG. 43 is a perspective view of a medical device according to an example implementation.
- (41) FIG. 44 is a top view of the medical device of FIG. 43.
- (42) FIG. 45 is a front view of the medical device of FIG. 43.
- (43) FIG. 46 is a side view of the medical device of FIG. 43.
- (44) FIG. 47 is an exploded top view of a barrel of a medical device according to an example implementation.
- (45) FIG. 48 is a flow chart illustrating an exemplary method including the medical device of FIG. 1.
- (46) FIG. 49 is a perspective view of a medical device according to one implementation.
- (47) FIG. 50 is an exploded view of the medical device of FIG. 49 with the locking plate removed according to one implementation.
- (48) FIG. 51 is a side view of the medical device of FIG. 50.
- (49) FIG. 52 is a front view of the medical device of FIG. 50.
- (50) FIG. 53 is a rear view of the medical device of FIG. 50.
- (51) FIG. 54 is a top view of the medical device of FIG. 50.
- (52) FIG. 55 is a side view of the medical device of FIG. 49 with the locking plate removed according to one example implementation.
- (53) FIG. 56 is a front view of the medical device of FIG. 55.
- (54) FIG. 57 is a rear view of the medical device of FIG. 55.
- (55) FIG. 58 is a top view of the medical device of FIG. 55.

- (56) FIGS. **59-62** illustrate assembly of an expandable central barrel of a medical device according to one example implementation.
- (57) FIGS. **63-65** illustrate assembly of the spikes for the medical device of FIGS. **59-62**.
- (58) FIG. **66** is a side view of a locking plate for a medical device according to one example implementation.
- (59) FIG. **67** is a front view of the locking plate of FIG. **66**.
- (60) FIG. **68** is a cross-sectional view of a trunion assembly for a medical device according to one example implementation.
- (61) FIG. **69** is a front view of the trunion assembly of FIG. **68**.
- (62) FIG. **70** is cross-sectional view of a tube that can be used to release the pawls of the trunion assembly of FIG. **68** according to one example implementation.
- (63) FIGS. **71-73** illustrate assembly of a locking plate for a medical device according to one example implementation.
- (64) FIG. **74** is a side view showing rotation of a locking plate for a medical device according to one example implantation.
- (65) FIG. **75** is a perspective view showing angulation of the spike assembly of the locking plate of FIG. **74**.
- (66) FIG. **76** is an end view of a medical device according to one example implementation.
- (67) FIG. **77** is a side view of the medical device of FIG. **76**.
- (68) FIG. **78** is a top view of the medical device of FIG. **76**.
- (69) FIG. **79** is an end view of the medical device of FIG. **76** with a locking plate removed in accordance with one example implementation.
- (70) FIG. **80** is a side view of the medical view of FIG. **76** with a locking plate removed in accordance with one example implementation.
- (71) FIG. **81** is a top view of the medical view of FIG. **76** with a locking plate removed in accordance with one example implementation.
- (72) FIGS. **82-85** illustrate assembly of the barrel for the medical device of FIG. **76** in accordance with one example implementation.
- (73) FIGS. **86-90** illustrate assembly of the spike assembly for the medical device of FIG. **76** in accordance with one example implementation.
- (74) FIGS. **91** and **92** illustrate the spike assembly of the medical device of FIG. **76** in accordance with one example implementation.
- (75) FIGS. **93** and **94** illustrate the locking plate of the medical device of FIG. **76** in accordance with one example implementation.
- (76) FIGS. **95** and **96** illustrate the trunion assembly for use with a locking plate in accordance with one example implementation.
- (77) FIG. **97** illustrates a tube that can be used with the trunion assembly in accordance with one example implementation.
- (78) FIGS. **98-100** illustrate assembly of a locking plate in accordance with one example implementation.
- (79) FIG. **101** illustrates rotation of a locking plate in accordance with one example implementation.

DETAILED DESCRIPTION

(80) Detailed implementations of the present invention are disclosed herein; however, it is to be understood that the disclosed implementations are merely examples of the invention, which may be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present invention in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting, but rather to provide an understandable description of the invention.

(81) The terms “a” or “an,” as used herein, are defined as one or more than one. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open transition).

(82) The devices and methods described herein are generally directed to medical devices that can be used to support, stabilize and/or replace anatomical structures within a body of a patient. In some implementations, the devices and methods described herein are configured to provide support to a spine or back of a patient, including providing support between two vertebrae in the spine or back of the patient. In other implementations, other portions of the body of the patient can be supported by the devices described herein.

(83) The medical devices described herein may be implanted within a body of a patient to assist in maintaining normal physiologic motion in the spine of the patient.

(84) The term patient may be used hereafter for a person who benefits from the medical device or the methods disclosed in the present invention. For example, the patient may be a person whose body receives the medical device disclosed by the present invention in a surgical treatment. For example, in some embodiments, the patient may be a human female, human male, or any other mammal.

(85) This document describes implementations of an implantable medical device that may be used as a posterior, non-pedicle supplemental fixation device for use in the non-cervical spine. The medical device may be used as an interspinous fusion device. The medical device may be implanted with or without the removal of the supraspinous ligament. In one or more implementations, the supraspinous ligament may be preserved. The medical device may be attached firmly to the spinous processes above and below an interspinous space. The medical device may immobilize a lumbar motion segment posteriorly with no other devices implanted. The medical device may withstand compressive, torsional and shear loads seen in the lumbar spine. The medical device may be used to achieve supplemental fusion and to treat conditions of the spine such as, for example, degenerative disc disease, spondylolisthesis, trauma (i.e., fracture or dislocation), tumor and/or other conditions.

(86) This document describes implementations of an implantable medical device, where the medical device may include an expandable central barrel with polyetheretherketone (PEEK) bone contacting endplates, with two spiked plates attached to the central barrel. For example, the two spiked plate may be held together on posterior rails. By way of further example, one of the spiked plates may be on one end of the expandable central barrel (e.g., integrally formed with the central barrel) with another one of the spiked plates being attached after the barrel is inserted into the interspinous space to clamp the device in place. The plates may include projections (e.g., spikes) that bite into the spinous process to clamp the device in place. Each of the plates may angulate to conform to the patient anatomy. The plates may be locked with a set screw and may have a lordotic profile to match the lumbar anatomy. The expandable barrel may provide interspinous distraction, off-loading the spikes on the plate and reducing the chances of breaking the spinous process. The barrel may be sized to fit into the interspinous space without resistance, and then expanded. The barrel may include a graft window (e.g., anteriorly and posteriorly) which may be packed with graft material after expansion. In some embodiments, the barrel includes a graft window anteriorly and posteriorly and can be packed with the graft material using the posterior window. The PEEK endplates may include anatomically-shaped grooves for optimal bone contact and fit.

(87) FIGS. 1-4 illustrate a medical device **10** according to one example implementation. The medical device **10** may be implanted in a patient and referred to as a spinous process fusion device. FIG. 1 is a perspective view of the medical device **10** with a barrel illustrated in a collapsed or contracted position and the plates in a separated position relative to one another. FIGS. 2-4 illustrate a top view, front view and side view, respectively, of the medical device **10** of FIG. 1, which illustrates the barrel in the collapsed or contracted position.

(88) The medical device **10** includes a first plate **12**, a second plate **14** and an expandable central

barrel (also referred to as a barrel) **16**. The barrel **16** is illustrated in a collapsed state. The barrel **16** includes a first portion **18** (e.g., a first rail **18**) and a second portion **20** (e.g., a second rail **20**). The first rail **18** and the second rail **20** also may be referred to as the rails **18** and **20**. The first rail **18** and the second rail **20** may be integrally formed with the barrel **16**. The first rail **18** and the second rail **20** also may be referred to as posterior rails. The first plate **12** and the second plate **14** (also referred to as the plates **12** and **14**) may be secured to the barrel **16** by coupling the first plate **12** to the first rail **18** and the second plate **14** to the second rail **20**. The first plate **12** and the second plate **14** each may include a bushing **22** (e.g., a spherical bushing) assembled into the plates **12** and **14**, where the plates **12** and **14** slide on the respective rails **18** and **20** through the bushing **22** and are secured using a set screw **24**. As discussed in more detail below, each plate **12** and **14** may move laterally along its respective rail **18** and **20** to engage spinous processes of adjacent vertebra above and below the interspinous space. FIGS. **1-4** illustrate the plates **12** and **14** in a separated position with respect to one another. Also, as discussed in more detail below, each plate **12** and **14** may angulate through a range of degrees with respect to the rails **18** and **20** to better conform to patient anatomy when implanted in a patient.

(89) In other example implementations (not shown), the first portion **18** and the second portion **20** may be grooves on the barrel **16**. In this example, the first plate **12** and the second plate **14** each may include a projection (e.g., a rail) that is movably inserted into the corresponding groove on the barrel **16**. This example implementation may function in the same way as described above and below, other than the structure of the rails may be implemented on the plates **12** and **14**, which are then received in the first portion **18** and the second portion **20** of the barrel **16**, where the first portion **18** and the second portion **20** are grooves on the barrel **16**.

(90) The first plate **12** may include an upper portion **26** and a lower portion **28**. The second plate **14** may include an upper portion **30** and a lower portion **32**. The plates **12** and **14** may include multiple projections **34** (e.g., spikes) on both the upper portions **26** and **30** and the lower portions **28** and **32**. While the term spikes may be used for the projections **34** other types of projections may be used that may have a more tapered point or rounded point or other type of ending to the projection. The spikes **34** may be used to attach firmly and bite into the spinous processes above and below an interspinous space. The spikes **34** may be integrally formed with the plates **12** and **14** or the spikes **34** may be separate components that are secured to the plates **12** and **14**. The spikes **34** may be pyramid shaped with a base portion secured or integrally formed on the plates **12** and **14**. The sides of the spikes **34** may extend from the base to form a point in the shape of a pyramid. In other example implementations, the spikes **34** may be formed into other shapes that rise to a point to enable the spike to engage the spinous process. As discussed above, the end of the spikes **34** may include tips other than a point such as, for example, rounded tip, a square tip or other-shaped tip.

(91) The plates **12** and **14** and the spikes **34** may be made of titanium. In other implementations, the plates **12** and **14** and the spikes **34** may be made of other biocompatible materials.

(92) The example illustration of the medical device **10** includes four (4) spikes **34** on each portion **26**, **28**, **30** and **32** of the plates **12** and **14**. In other example implementations, fewer or more spikes **34** may be included. In one example implementation, the spikes **34** on opposing portions (i.e., upper portions **26** and **30** and lower portions **28** and **32**) may be aligned across from one another. In other example implementations, the spikes **34** on opposing portions may be offset from one another.

(93) The first plate **12** and the second plate **14** may be shaped in a lordotic profile to match the lumbar anatomy. With respect to the first plate **12**, the upper portion **26** is connected to the lower portion **28** by a central portion **36**. The upper portion **26**, the lower portion **28** and the central portion **36** may be integrally formed as a single plate component. The central portion **36** includes an open side (e.g., a C-shaped opening) to receive the bushing **22** and an opening (e.g., a hole) to receive the set screw **24**, as illustrated in more detail in FIGS. **24-27**. In other example implementations, the first plate **12** and the second plate **14** may be other shapes suitable for a

particular application.

(94) Similarly to the first plate **12**, the second plate **14** includes a central portion **38** that connects the upper portion **30** to the lower portion **32**. The upper portion **30**, the lower portion **32** and the central portion **38** may be integrally formed as a single plate component. The central portion **38** include an open side (e.g., a C-shaped opening) to receive the bushing **22** and an opening (e.g., a hole) to receive the set screw **24**, as illustrated in more detail in FIGS. **24-27**. The set screw **24** is used to lock the plates **12** and **14** in an angular position at any position within their range of angular motion.

(95) The central barrel **16** is an expandable barrel that may be in a collapsed position for insertion into a patient in the interspinous space without resistance and then expanded up to the barrel's maximum height. In one example implementation, the maximum expanded height of the barrel may be about 4 mm greater than the collapsed height.

(96) The central barrel **16** includes a first endplate **40** and a second endplate **42** (also referred to as endplates **40** and **42**), as best viewed in FIG. **3**. Each of the endplates **40** and **42** includes a respective groove **44** and **46**. The grooves **44** and **46** may be anatomically-shaped grooves optimal bone contact and fit in the patient. The endplates **40** and **42** may be PEEK bone contacting endplates. The barrel **16** may be bullet-shaped on both ends in the lateral and posterior directions to facilitate insertion into a patient. The expandable barrel **16** may provide interspinous distraction and may offload the forces of the spikes **34** on the plates **12** and **14** to reduce the chances of breaking a spinous process. The barrel **16** may be inserted, laterally or posteriorly, in a smaller height and then expanded to provide distraction, eliminating forces on the spinous process and potential frustration for a surgeon performing the implantation.

(97) The barrel **16** includes a first window **48** (e.g., also referred to as an opening or an anterior window) and a second window **50** (e.g., also referred to as an opening or a posterior window). The first window **48** and the second window **50** may be used as graft windows for the packing of bone graft material following the insertion and placement of the medical device **10** in a patient. In one implementation, after the barrel **16** has been expanded, the barrel **16** may be packed with bone graft using the second window **50**. In this manner, graft containment areas accessed by the windows **48** and **50** may provide for a larger grafting area and may be packed after expansion of the barrel **16**.

(98) Referring to FIGS. **5-8**, an example implementation of the medical device **10** of FIGS. **1-4** is illustrated with the barrel **16** shown in an expanded state and the plates **12** and **14** shown in a separated position with respect to one another. That is, the plates **12** and **14** are each positioned towards an outer end of the rails **18** and **20**. The barrel **16** expands and contracts by expanding and contracting the endplates **40** and **42** in a direction towards the upper **26** and **30** and lower portions **28** and **32** of the sides **12** and **14**, respectively. The mechanism to expand and contract the barrel **16** is illustrated in more detail in FIGS. **28-30** below.

(99) In general, a central screw **52** is rotated to actuate two independent internal actuators. The actuators include split ramps that raise and lower the endplates **40** and **42** when the central screw **52** is rotated. FIGS. **7** and **8** provide views that illustrate the barrel **16** in a fully expanded state. As discussed above, the barrel **16** may be expanded after insertion into the interspinous space. After expansion, the barrel **16** may be packed with bone graft material using the window **50**. Prior to expansion, some bone graft material may be packed into the barrel **16** using the window **48**.

(100) Referring to FIGS. **9-12**, an example implementation of the medical device **10** of FIGS. **1-4** is illustrated with the barrel **16** shown in a collapsed state and the plates **12** and **14** shown in a closed position. That is, the plates **12** and **14** have been traversed along the rails **18** and **20** towards one another. The plates **12** and **14** may slide along the rails **18** and **20** and may be secured in position at any point along the rails **18** and **20** using the set screw **24**. When the plates **12** and **14** are slid together along the rails **18** and **20**, the spikes **34** on the plates **12** and **14** may engage and clamp (or bite into) the spinous process. In this manner, the spikes **34** on the upper portions **26** and **30** may clamp together and into one spinous process and the spikes **34** on the lower portions **28** and

32 may clamp together and into an adjacent spinout process.

(101) As illustrated in FIGS. **9-12**, the spikes **34** on one plate are aligned to mate at a same point with the spikes **34** on an opposing plate. In other example implementations, the spikes **34** on one plate may be offset in relation to the spikes **34** on an opposing plate.

(102) Referring to FIGS. **13-16**, an example implementation of the medical device **10** of FIGS. **1-4** is illustrated with the barrel **16** shown in an expanded state and the plates **12** and **14** shown in a closed position. In this manner, this illustrates the medical device **10** in a state after insertion into the patient such that the plates **12** and **14** have been traversed along the rails **18** and **20** to clamp on the spinous process of adjacent vertebrae and the barrel **16** has been expanded using the central screw **52**.

(103) Referring to FIGS. **17-23**, an example implementation of the medical device **10** of FIGS. **1-4** is illustrated with the barrel **16** shown in an expanded state and the plates **12** and **14** shown in an open or separated position and in an angulated configuration. As discussed above, the plates **12** and **14** may rotate angularly with respect to the rails **18** and **20**. The plates **12** and **14** may pivot around the bushing **22** and may be locked in place using the set screw **24**. In one example implementation, the plates **12** and **14** may have a range of motion of about 25 degrees offset with respect to the rails **18** and **20**. The angulation of the plates **12** and **14** enables each plate to conform independently to the anatomy of the particular patient. Each plate **12** and **14** may be pivoted and locked at any position in their range of motion independent of the other plate.

(104) In FIGS. **22** and **23**, a side view (FIG. **22**) and a detailed view of inset A (FIG. **23**) illustrate that the plates **12** and **14** are locked using the set screw **24**. The rails **18** and **20** may be C-shaped or curved and include a groove area **60**. The set screw **24** may include a curved, cup-shaped design on the tip **62**. The curved tip **62** penetrates through the opening in the rail **14** and through the bushing **22** to engage the groove area **62** of the rail **20** to secure and lock the plate **14** in place. The curved tip **62** maximizes the surface contact with the groove area **62** of the rail **20** when the plate **14** pivots through its range of motion. FIGS. **24-27** below also illustrate the curved (or cup-shaped or bulleted) tip **62** of the set screw **24**.

(105) Referring to FIGS. **24-27**, the assembly of the plates **12** and **14** is illustrated. In these example figures, plate **14** is referenced for illustrative purposes. The plate **14** may be assembled by placing the bushing **22** into the plate initially offset by 90 degrees from its final position. As described above, the bushing **22** may be a spherical bushing that is shaped to be positioned on and traverse the rail **20** on the barrel **16**. The bushing **22** may include a slot **64** or opening in the back of the bushing to receive the set screw **24**.

(106) Once the bushing **22** has been inserted into the plate **14** (FIG. **25**), the bushing **22** is rotated 90 degrees into its final position in the plate **14** (FIG. **26**). Then, the set screw **24** having the curved tip **62** may be inserted through the opening in the back of the plate **14** through the slot **64** in the bushing **22**. The set screw **24** serves to prevent the bushing **22** from rotating back out of the plate **14**.

(107) Referring to FIGS. **28-30**, the barrel **16** and assembly of the barrel **16** is illustrated in detail. As discussed above, the barrel **16** includes a first endplate **40** and a second endplate **42**. The endplates **40** and **42** may be PEEK endplates. The barrel **16** includes a central screw **52** having a first thread portion **66** and a second thread portion **68**. The barrel **16** includes a frame **65**, a first actuator **70** and a second actuator **72** (also referred to as the actuators **70** and **72**) and two assembly pins (not shown). In one example implementation, the frame **65**, the actuators **70** and **72** and the central screw **52** may be made of titanium. In other example implementations, the components may be made of other biocompatible materials.

(108) Each of the actuators **70** and **72** may include split ramps **74** and **76** to accommodate the curved shape of the barrel **16**. The barrel **16** is curved shaped and may be bulleted (or egg-shaped) on each end to allow for easier insertion into the interspinous space. The curved shape of the barrel **16** may provide maximum graft packing volume.

(109) The actuators **70** and **72** may be loosely assembled into the frame **65** of the barrel **16** and the **74** and **76** placed over the actuators **70** and **72**. The central screw **52** may be inserted into the actuators **70** and **72** and timed so that the actuators have specific spacing per rotation of the screw **52**. Once the screw **52** is fully inserted, two pins (not shown) are pressed into the frame **65** posteriorly to capture the screw **52** to prevent its disassembly.

(110) The rotation of the screw **52** causes the actuators **70** and **72** to rotate and the ramps **74** and **76** on the actuators **70** and **72** to push against the endplates **40** and **42**, causing the endplates **40** and **42** to expand from a collapsed position. A counter rotation of the screw **52** causes the actuators **70** and **72** to rotate and the ramps **74** and **76** on the actuators **70** and **72** to recede from pushing against the endplates **40** and **42**, causing the endplates **40** and **42** to collapse from an expanded state.

(111) FIGS. **31-34** illustrate a medical device **100** according to an example implementation. Similarly to the medical device **10**, the medical device **100** may be implanted in a patient and referred to as a spinous process fusion device. Like reference numbers between the FIGS. **1-30** and FIGS. **31-34**, and other figures below describing medical device **100**, refer to the same or similar components and features between the two medical devices. The medical device **100** may have the same features and functionality as the medical device **10**.

(112) The medical device **100** includes a first plate **12** and a second plate **14**. The medical device **100** includes a barrel **116**. In the example of FIGS. **31-34**, the barrel **116** includes rails **118** and **120** that each extend from a different side of the barrel **116** instead of extending from a same side like the rails **18** and **20** from the barrel **16** in medical device **10**. The barrel **116** is essentially rotated 90 degrees compared to the barrel **16**. In other aspects, the barrel **116** is an expandable barrel and has the same functionality as the barrel **16**. The barrel **116** may be inserted laterally into a patient in the interspinous space. The barrel **116** may be inserted at a smaller height (or in a collapsed state) and then expanded to provide distraction and to eliminate the forces on the spinous process and frustration for the surgeon.

(113) In FIGS. **31-34**, the medical device **100** illustrates the plates **12** and **14** in an open state and the barrel **116** in a collapsed state. In this manner, the medical device **100** may be inserted into a patient and then the barrel **116** expanded.

(114) Referring to FIGS. **35-38**, the medical device **100** is illustrated with the barrel **116** in an expanded state. In one example implementation, the expanded barrel height for the barrel **116** may be about 7 mm greater than the collapsed height. The sides **12** and **14** are illustrated in an open state. The barrel **116** may be expanded from a collapsed state to an expanded state using the central screw **152**. Similarly, the barrel **116** may be collapsed from an expanded state to a collapsed state using the central screw **152**.

(115) Referring to FIGS. **39-42**, the medical device **100** is illustrated with the barrel **116** in an expanded state and the plates **12** and **14** in a closed position. As discussed above with respect to the medical device **10**, the plates **12** and **14** on the medical device **100** also may traverse the rails **118** and **120** of the barrel between an open position and a closed position. In the closed position, the plates **12** and **14** are designed to clamp and bite into the spinous process, as discussed above in detail.

(116) Referring to FIGS. **43-46**, the medical device **100** is illustrated with the barrel **116** in an expanded state and the plates **12** and **14** in a closed and angulated position. As discussed above with respect to FIGS. **17-20**, the plates **12** and **14** may angulate about 25 degrees with respect to the rails **118** and **120** to better conform to patient anatomy. The plates **12** and **14** may be locked in position using the set screw **24**.

(117) Referring to FIG. **47**, the barrel **116** is assembled in a manner similar to the barrel **16**, as discussed above with respect to FIGS. **28-30**. The barrel **116** includes a first endplate **140** and a second endplate **142**, two independent actuators with ramps and a central screw **152**. The endplates **140** and **142** are loosely assembled into the actuator ramps and the central screw **152** is inserted into the actuator ramps, which anchor the assembly together.

(118) Referring to FIG. 48, an example flowchart illustrates an example process 200 for using the medical devices 10 and 100. For example, process 200 includes inserting a barrel 16 or 116 of the medical device 10 or 100, respectively, into an interspinous space (210). As discussed above, the medical device includes the barrel 16 or 116 having a first portion (e.g., rail 18 or 118) and a second portion (e.g., rail 20 or 120), a first plate 12 having multiple projections 34 extending from one side of the first plate 12 and a second plate 14 having multiple projections 34 extending from one side of the second plate (210).

(119) The process 200 includes expanding the barrel 16 or 116 from a collapsed form having a first height to an expanded form having a second height, where the second height is greater than the first height (220). As discussed above, the central screw 52 or 152 may be rotated to expand the barrel 16 or 116 from a collapsed form to an expanded form in the interspinous space.

(120) The process includes moving the first plate 12 on the first portion (e.g., rail 18 or 118) to engage a spinous process (230) and moving the second plate 14 on the second portion (e.g., rail 20 or 120) to engage the spinous process (240). For example, the projections 34 on each of the plates 12 and 14 may engage the spinous process of adjacent vertebrae as the plates 12 and 14 are slid along the respective rails.

(121) Optionally, the process 200 may include positioning the first plate 12 to a desired angle with respect to the first portion and positioning the second plate 14 to a desired angle with respect to the second portion. Once the plates 12 and 14 have been positioned to their desired angles, the plates 12 and 14 may be locked into position using the set screws 24.

(122) FIG. 49 illustrates a medical device 300 according to one example implementation. FIG. 49 is a perspective view of the medical device 300. The medical device 300 may be implanted into a patient and referred to as a spinous process fusion device. In the illustrated embodiment, the medical device 300 includes a first plate 302, a second plate 304 (e.g., locking plate 304), and an expandable central barrel (also referred to as a barrel) 306.

(123) With additional reference to FIGS. 50-54, the medical device 300 of FIG. 49 will be described in more detail. FIGS. 50-54 illustrate an exploded view, side view, front view, rear view, and top view, respectively, of the medical device 300, which illustrate the barrel 306 in the collapsed or contracted position with the locking plate 304 removed. The barrel 306 may be inserted into the interspinous space without the locking plate 304 and then expanded. The locking plate 304 may then be attached to the barrel 306 after insertion to lock the medical device 300 in place in engagement with the spinous process.

(124) In the illustrated embodiment, the barrel 306 includes a first portion 308 (e.g., upper portion 308) and a second portion 310 (e.g., lower portion 310). The first portion 308 may include a pair of ramped upper sidewalls 312. The ramped upper sidewalls 312 may include ramped portions 316 on either end of the ramped sidewalls 314. The second portion 312 may also include a pair of ramped lower sidewalls 314. The ramped lower sidewalls 314 may include ramped portions 318 on either end. As best seen in FIG. 51, the ramped lower sidewalls 314 and the ramped upper sidewalls 312 may overlap when the medical device 10 is collapsed. The ramped upper sidewalls 312 and the ramped lower sidewalls 314 may define a central chamber in the barrel 306. The central chamber 315 may be used for the packing of bone graft material following the insertion and placement of the medical device 10 in a patient. In one implementation, after the barrel 306 has been expanded, the barrel 306 may be packed with bone graft using the central chamber 315. In this manner, the central chamber 315 may provide for a larger grafting area and may be packed after expansion of the barrel 306.

(125) The central barrel 306 is an expandable barrel that may be in a collapsed position for insertion into a patient in the interspinous space without resistance and then expanded up to the barrel's maximum height. In one example implementation, the maximum expanded height of the barrel may be about 4 mm greater than the collapsed height or, alternatively, about 6 mm greater than the collapsed height. The central barrel 306 may provide interspinous distraction and may

offload the forces of the spikes **328**, **362** on the plates **302** and **304** to reduce the chances of breaking a spinous process. The barrel **306** may be inserted, laterally or posteriorly, in a smaller height and then expanded to provide distraction, eliminating forces on the spinous process and potential frustration for a surgeon performing the implantation.

(126) The first plate **302** may include an upper portion **320** and a lower portion **322**. The upper portion **320** of the first plate **302** may extend generally vertically from the first portion **308** of the barrel **306**. The upper portion **320** may be integrally formed with the first portion **308**. The lower portion **322** of the first plate **302** may extend from the second portion **310** of the barrel **306** in a direction generally opposite to the upper portion **320**. The lower portion **322** may be integrally formed with the second portion **310**. The first plate **302** may be shaped in a lordotic profile to match the lumbar anatomy.

(127) The first plate **302** may include a spike assembly **324** on both the upper portion **320** and the lower portion **322**. The spike assemblies **324** may each be received within an opening **326** in both the upper portion **320** and the lower portion **322**. Each spike assembly **324** may include multiple projections (e.g., spikes **328**) that extend from a spike sphere **330**. The spike spheres **330** may each be a complete sphere, hemisphere, or a spheric section. Each spike assembly **324** may further comprise a wedge **332** and a post **334**. The wedge **332** may be secured onto the post **334** with the spike sphere **330** fit onto the wedge **332** over the post **334**. A pin (not shown) may be used in the opening **326** to prevent rotation of the spike sphere **330** in the opening **326** while allowing articulation of the spike sphere **330** with respect to the first plate **302**. Slots **336** may be disposed in the spike sphere **330**, as best seen on FIG. 49.

(128) While the term “spikes” may be used for the projections other types of projections may be used that may have a more tapered point or rounded point or other type of ending to the projection. The spikes **328** may be used to attach firmly and bite into the spinous processes above and below an interspinous space. While spike assemblies **324** are shown, other embodiments may include spikes **328** that are integrally formed with the first plate **302**. The spikes **328** may be pyramid shaped with a base portion secured or integrally formed on the spike sphere **330**. The sides of the spikes **328** may extend from the base to form a point in the shape of a pyramid. In other example implementations, the spikes **328** may be formed into other shapes that rise to a point to enable the spike to engage the spinous process. As discussed above, the end of the spikes **328** may include tips other than a point such as, for example, rounded tip, a square tip or other-shaped tip. The example illustration of the medical device **10** includes three (3) spikes **328** on each spike assembly **324** of the first plate **302**. In other example implementations, fewer or more spikes **328** may be included. The first plate **302** and the spikes **328** may be made of titanium. In other implementations, the first plate **302** and the spikes **328** may be made of other biocompatible materials.

(129) The medical device **10** may further include an actuator assembly **338** (best seen on FIG. 50) for raising and lowering the first and second portions **308** and **310** of the barrel **306** and, thus, the upper and lower portions **320**, **322** of the first plate **302**. The actuator assembly **338** may be disposed between the first and second portions **308** and **310** of the barrel **306**. As illustrated, the actuator assembly **338** may comprise a central screw **340**, a front ramped actuator **342** and a rear ramped actuator **344**. The front ramped actuator **342** may be bullet shaped on its front end to facilitate insertion into a patient. The front ramped actuator **342** may have a ramped expansion portion **346** and an extension portion **348**. The ramped expansion portion **346** may be located at a front end of the barrel **306** with the extension portion **348** extending from the ramped expansion portion **346** towards a rear end of the barrel **306**. The central screw **340** may extend through the barrel **306** and engage the extension portion **348**. The first and second portions **308** and **310** of the barrel **306** may slidably engage the ramped expansion portion **346**. For example, the ramped expansion portion **346** may engage ramped surface **316** of the first and second portions **308** and **310** at a front end of the barrel **306**. The ramped expansion portion **346** may have dovetail connections with the first and second portions **308** and **310**, respectively. The rear ramped actuator

344 may be disposed at a rear end of the barrel **306**. The first and second portions **308** and **310** of the barrel **306** may slidably engage the rear ramped actuator **344**. For example, the rear ramped actuator **344** may also engage ramped surfaces **316** of the first and second portions **308** and **310** of the barrel. The rear ramped actuator **344** may have dovetail connections with the first and second portions **308** and **310**, respectively. The central screw **340** may extend through the rear ramped actuator **344** to engage the extension portion **348**.

(130) Referring to FIGS. **56-58**, an example implementation of the medical device of FIGS. **49-55** is illustrated with the barrel **306** shown in an expanded state. The barrel **306** expands by forcing the first and second portions **308** and **310** vertically outward in a direction away from one another. In this manner, the upper and lower portions **320** and **322** of the first plate **302** are also expanded vertically outward. The barrel **306** contracts by forcing the first and second portions **308** and **310** to contract in a direction toward one another, thus also moving the upper and lower portions **320** and **322** of the first plate **302** together. In some embodiments, the actuator assembly **338** may be used to raise and lower the first and second portions **308** and **310**. By way of example, the central screw **340** may be turned to contract the actuator assembly **338**. The rear ramped actuator **342** may be held in place while the central screw **340** is turned causing the front ramped actuator **340** to be drawn toward the rear ramped actuator **342**. The rear ramped actuator **342** and the front ramped actuator **340** may engage the ramped upper sidewalls **312** and the ramped lower sidewalls **314** in the first and second portions **308** and **310** forcing the first and second portions **308** and **310** to expand from a collapsed position. A counter rotation of the central screw **340** may cause the front ramped actuator **340** and the rear ramped actuator **342** to separate causing the first and second portions **308** and **310** to collapse from the expanded state.

(131) Referring to FIGS. **59-62**, assembly of the barrel **306** of the medical device **300** shown on FIGS. **49-58** will now be described according to an example implementation. As illustrated by FIG. **59**, the barrel **306** may comprise a first portion **308** and a second portion **310**. The first plate **302** may be defined by upper portion **320** and lower portion **322**. Upper portion **320** may extend from first portion **308** of the barrel **306**, and lower portion **322** may extend in an opposite direction from second portion **310** of the barrel **306**. As further illustrated by FIG. **59**, the actuator assembly **338** may comprise a central screw **340**, a front ramped actuator **342**, and a rear ramped actuator **344**. In FIG. **60**, the rear ramped actuator **344** may be slid onto the first portion **308** of the barrel **306**. As illustrated, the rear ramped actuator **344** may engage (e.g., through a dovetail connection) a rear end of the upper ramped sidewalls **312** of the first portion **308**. In FIG. **61**, the front ramped actuator **342** may then slide onto the second portion **310** of the barrel **306**. As illustrated, the ramped expansion portion **346** may engage (e.g., through a dovetail connection) a front end of the lower ramped sidewalls **314** of the second portion **310**. In FIG. **62**, the first portion **308** and second portion **310** of the barrel **306** have been placed together in a contracted position with a front end of the upper ramped sidewalls **312** engaging the ramped expansion portion **346** and a rear end of the lower ramped sidewalls **314** engaging the rear ramped actuator **344**.

(132) Referring to FIGS. **63-65**, assembly of the spike assemblies **324** of the medical device **300** shown on FIGS. **49-58** will now be described according to an example implementation. As illustrated by FIG. **63**, the spike assemblies **324** each comprise a spike sphere **330**, a wedge **332**, and a post **334**. In FIG. **64**, the post **334** may be inserted into the wedge **332** coupling the post **334** and the wedge **332**. The wedge **332** may be secured onto one end of the post **334**. Each assembly of the post **334** and wedge **332** may then be placed into the opening **326** in the upper and lower portions **320** and **322** of the first plate **302**. The spike sphere **330** may then be placed onto the other end of the post **334**, which may be then pressed back into the opening **326**, as seen in FIG. **65**. In one embodiment, a snap connection may secure the spike sphere **330** to the post **334**. A pin (not shown) may be used in the opening **326** to prevent rotation of the spike sphere **330** assembly **324** in the opening **326** while allowing articulation of the spike sphere **330** with respect to the first plate **302**. In some embodiments, slots **336** in the spike sphere **330** allow the spike sphere **330** to expand and

collapse. To lock the spike sphere **330** in a particular orientation, the wedge **332** may be compressed further into the opening **326** causing the spike sphere **330** to expand outward and lock. (133) Referring to FIGS. **49** and **66-67**, the second or locking plate **304** will now be described in more detail with respect to one example implementation. FIG. **49** is a perspective view of the medical device **10** with the locking plate **304**. FIGS. **66** and **67** are side and front views, respectively, of the locking plate **304**. The locking plate **304** may be inserted onto the central screw **340** after the barrel **306** has been expanded to lock the barrel **306** in position.

(134) As illustrated, the locking plate **304** may comprise an upper portion **350** and a lower portion **352**. A central portion **354** may connection the upper portion **350** to the lower portion **352**. The upper portion **350**, lower portion **352**, and central portion **354** may be integrally formed as a single plate component. The central portion **354** includes an opening (e.g., a central opening) to receive trunion assembly **356** (best seen on FIG. **67**). The locking plate **304** may rotate about the trunion assembly **356** and can be locked at various angles at any position within its range of motion. In some embodiments, the trunion assembly **356** may be configured so that the locking plate **304** rotates about its center. The locking plate **304** may include a spike assembly **358** on both the upper portion **350** and the lower portion **352**. The spike assemblies **358** may each be received within an opening in both the upper portion **350** and the lower portion **352** of the locking plate **304**. Each spike assembly **358** may comprise a spike sphere **360** having multiple projections, such as spikes **362**. Each spike assembly **358** may further comprise a wedge **364** and a post **366**. The spike assemblies **358** and its various components may be similar in function and assembly to the spike assembly **324** of the first plate **302** discussed above with respect to FIGS. **49-58** and **63-65**.

(135) With additional reference to FIGS. **68** and **69**, the trunion assembly **356** will described in more detail with respect to one example implementation. As illustrated, the trunion assembly **356** may comprise a housing **368**. The housing **368** may have laterally extending projections **370** for rotatably coupling the trunion assembly **368** to the central portion **354** of the locking plate **304** while allowing the locking plate **304** to rotate with respect to the trunion assembly **368**. As illustrated, there may be a pair of projections **370** that extend from opposite sides of the housing **368** and are each received in corresponding openings **372** in the central portion **354**. The housing **368** may further have a through bore **374** for receiving the central screw **340**. The housing **368** may further comprise a pair of chambers **376** on either side of the through bore **374**. The housing **368** may further include a ratchet pawl **378** in each chamber **376**. Embodiments of the ratchet pawls **378** may be spring loaded so that the ratchet pawls **378** may maintain contact with the central screw **340** while the locking plate **304** rotates about the trunion assembly **368**. The ratchet pawls **378** may be assembled from the side of the housing **368**. The ratchet pawls **378** may each have spring cuts to allow the ratchet pawls **378** to compress further into the chambers **376**. The spring cuts may be the height of an electric discharge machining wire to create a small gap within each leaf of the ratchet pawls **378** being self-limiting as it collapses upon itself. Insertion of the central screw **340** into the through bore **374** (e.g., from right to left of FIG. **68**) should cause the teeth (or threading) of the central screw **340** to engage the ratchet pawls **378** causing the ratchet pawls **378** to recess into the chambers **376**. The angling of the teeth on the ratchet pawls **378** should resist backwards motion of the central screw **340** after insertion into the through bore **374**. In this manner, the ratchet pawls **378** may be operable to secure the trunion assembly **368** and thus the locking plate **304** onto the central screw **340**.

(136) FIG. **70** illustrates a tube **380** that can be used to release the ratchet pawls **378** in accordance one example implantation. The tube **380** may be sized to fit over the central screw **340**. The tube **380** may be advanced over the central screw **340** and into the back end of the through bore **374** until the leading end or nose **382** of the tube **380** engages the ratchet pawls **378**. Pressure from the tube **380** combined with large chamfers on the ratchet pawls should cause the ratchet pawls **378** to compress. When fully inserted, the tube **380** includes one or more teeth **384** configured to snap into the ratchet pawls **378** allowing complete release of the central screw **340**.

(137) As illustrated by FIGS. 68 and 69, the housing 368 may have an upper surface 386 and a lower surface 388. In embodiments, the upper and lower surfaces 386 and 388 may each be curved. As illustrated, the upper and lower surfaces 386 and 388 may be sloped inward from the rear to the front of the housing 368. In some embodiments, the upper and lower surfaces 386 and 388 may each comprise a projection 390. The projection 390 may engage the locking plate 304 to limit its rotation about the trunion assembly 356.

(138) Referring to FIGS. 71-75, assembly of the locking plate 304 shown on FIGS. 66 and 67 will now be described according to an example implementation. As illustrated by FIG. 71, the locking plate 304 may comprise an upper portion 350, a lower portion 352, and a central portion 354 coupling the upper portion 350 and the lower portion 352. The trunion assembly 356 may comprise a housing 368 and a pair of ratchet pawls 378. The housing 368 may comprise a pair of windows 392 for receiving the ratchet pawls 378 into chambers 376 (FIG. 68). The ratchet pawls 378 may be inserted into the housing 368 from the side via windows 392, as shown on FIG. 72. The trunion assembly 356 comprising the housing 368 having the ratchet pawls 378 disposed therein may then be inserted into the opening in the central portion 354 of the locking plate 304, as best seen in FIG. 73. The trunion assembly may comprise projections 370 that are received in openings 372 in the central portion 354 to secure the trunion assembly 356 in the central portion 354. The projections 370 may be chamfered or otherwise angled on their leading edges to allow insertion into the openings 372. The plate assemblies 358 may then be inserted into the upper portion 350 and lower portion 352 of the locking plate 304. In embodiments, the spike assemblies 358 may be assembled and inserted into the locking plate 304 in a manner similar to that discussed above with respect to FIGS. 63-65.

(139) As previously mentioned, the locking plate 304 may be free to rotate about the trunion assembly 356 even where the trunion assembly 356 is in engagement with central screw 340. FIG. 74 is a side view of the locking plate 304 illustrating rotation of the locking plate 304 according to one example implementation. Additionally, the spike spheres 360 may also be free to articulate with respect to the locking plate 304. FIG. 74 illustrates articulation of the spike spheres 360 in accordance to one example implantation. Rotation of the locking plate 304 and/or articulation of the spike spheres 360 can provide an adaptable medical device 10 that can accommodate variances in spinous process geometry, for example, with the goal of anterior and secure placement.

(140) An embodiment for using the medical device 300 will now be described in accordance with one example implementation. For example, a method may comprise inserting the barrel 306 of the medical device 300 into an interspinous space. The method may further comprise expanding the barrel 306 from a collapsed form having a first height to an expanded form having a second height, where the second height is greater than the first height. As discussed above, the central screw 340 may be rotated to expand the barrel 306 from a collapsed form to an expanded form in the interspinous space. The process may further include inserting the locking plate 304 onto the central screw 340 and moving the locking plate 304 towards the first plate such that the locking plate 304 and the first plate 302 engage a spinous process. The locking plate 304 may be free to rotate about its center (e.g., the trunion assembly 356) to accommodate spinous process geometry. In addition, the spike spheres 324 and 360 of the first plate 302 and the locking plate 304, respectively, may also be free to articulate for accommodation of spinous process geometry. The spike spheres 324 and 360 may be locked into place during compression into the spinous process.

(141) FIGS. 76-78 illustrate a medical device 400 according to one example implementation. FIGS. 76-78 are an end view, side view, and top view, respectively, of the medical device 400. The medical device 400 may be implanted into a patient and referred to as a spinous process fusion device. In the illustrated embodiment, the medical device 400 includes a first plate 402, a second plate 404 (e.g., also referred to as locking plate 404), and an expandable central barrel (also referred to as a barrel) 406. The first plate 402 and the second plate 404 expand in overall height as the medical device 400 expands. The first plate 402 and the second plate 404 each include spikes

428, best seen on FIG. 77, which engage the spinous process.

(142) FIGS. 79-81 illustrate an end view, side view, and top view, respectively, of the medical device **400**, which illustrate the barrel **406** in the expanded position with the locking plate **404** removed. The locking plate **404** may be attached after the barrel **406** with the first plate **402** is inserted to lock the device **400** in place. The barrel **406** may be inserted into the interspinous space without the locking plate **404** and then expanded. The locking plate **404** may then be attached to the barrel **406** after insertion to lock the medical device **400** in place in engagement with the spinous process. As best seen on FIG. 78, the medical device **400** may include a central chamber **415** (or graft window) for the packing of bone graft material, which can be packed before or after insertion. The design of the medical device **400** may allow for placement with or without removal of the supraspinous ligament, depending on the surgeon's preference, for example.

(143) Referring now to FIGS. 82-85, the barrel **406** will now be described in more detail in accordance with present embodiments. FIGS. 82-85 illustrate an example technique for assembly of the barrel **406**. In the illustrated embodiment, the barrel **406** includes a first portion **408** (e.g., upper portion **408**) and a second portion **410** (e.g., lower portion **410**). The first portion **408** may include upper sidewalls **412**. The ramped upper sidewalls **412** may include ramped portions **416** on either end of the ramped sidewalls **414**. The second portion **412** may also include ramped lower sidewalls **414**. The ramped lower sidewalls **414** may include ramped portions **418** on either end. As best seen in FIG. 85, the ramped lower sidewalls **414** and the ramped upper sidewalls **412** may overlap when the medical device **400** is collapsed.

(144) The barrel **406** in the illustrated embodiments may be an expandable barrel that may be in a collapsed position for insertion into a patient in the interspinous space without resistance and then expanded up to the barrel's maximum height. In one example implementation, the maximum expanded height of the barrel may be about 4 mm greater than the collapsed height or, alternatively, about 6 mm greater than the collapsed height. The central barrel **406** may provide interspinous distraction and may offload the forces of the spikes **428** on the plates **402** and **404** to reduce the chances of breaking a spinous process. The barrel **406** may be inserted, laterally or posteriorly, in a smaller height and then expanded to provide distraction, eliminating forces on the spinous process and potential frustration for a surgeon performing the implantation.

(145) With continued reference to FIGS. 82-85, the first plate **402** may include an upper portion **420** and a lower portion **422**. The upper portion **420** of the first plate **402** may extend generally vertically from the first portion **408** of the barrel **406**. As illustrated, the upper portion **420** may extend from one end of the barrel **406**. The upper portion **420** may be integrally formed with the first portion **408**. The lower portion **422** of the first plate **402** may extend from the second portion **410** of the barrel **406** in a direction generally opposite to the upper portion **420**. As illustrated, the lower portion **420** may extend from the same end of the barrel **406** as the upper portion **420** but in the opposite direction. The lower portion **422** may be integrally formed with the second portion **410**. The first plate **402** may be shaped in a lordotic profile to match the lumbar anatomy.

(146) The medical device **400** may further include an actuator assembly **438** (best seen on FIGS. 82-85) for raising and lowering the first and second portions **408** and **410** of the barrel **406** and, thus, the upper and lower portions **420**, **422** of the first plate **402**. The actuator assembly **438** may be disposed between the first and second portions **408** and **410** of the barrel **406**. As illustrated, the actuator assembly **438** may comprise a central screw **440**, a front ramped actuator **446** and a rear ramped actuator **444**. The front ramped actuator **446** may be bullet shaped on its front end to facilitate insertion into a patient. The front ramped actuator **446** may have a ramped expansion portion **442** and an extension portion **448**. The ramped expansion portion **442** may be located at a front end of the barrel **406** with the extension portion **448** extending from the ramped expansion portion **442** towards a rear end of the barrel **406**. The central screw **440** may extend through the barrel **406** and engage the extension portion **448**. The first and second portions **408** and **410** of the barrel **406** may slidably engage the ramped expansion portion **442**. For example, the ramped

expansion portion **442** may engage ramped surface **416** of the first and second portions **408** and **410** at a front end of the barrel **406**. The ramped expansion portion **442** may have dovetail connections with the first and second portions **408** and **410**, respectively. The rear ramped actuator **444** may be disposed at a rear end of the barrel **406**. The first and second portions **408** and **410** of the barrel **406** may slidably engage the rear ramped actuator **444**. For example, the rear ramped actuator **444** may also engage ramped surfaces **416** of the first and second portions **408** and **410** of the barrel. The rear ramped actuator **444** may have dovetail connections with the first and second portions **408** and **410**, respectively. The central screw **440** may extend through the rear ramped actuator **444** to engage the extension portion **448**.

(147) An example embodiment for expanding the barrel **406** will now be described. The barrel **406** may expand by forcing the first and second portions **408** and **410** vertically outward in a direction away from one another. In this manner, the upper and lower portions **420** and **422** of the first plate **402** may also be expanded vertically outward. The barrel **406** may contract by forcing the first and second portions **408** and **410** to contract in a direction toward one another, thus also moving the upper and lower portions **420** and **422** of the first plate **402** together. In some embodiments, the actuator assembly **438** may be used to raise and lower the first and second portions **408** and **410**. By way of example, the central screw **440** may be turned to contract the actuator assembly **438**. The rear ramped actuator **442** may be held in place while the central screw **440** is turned causing the front ramped actuator **440** to be drawn toward the rear ramped actuator **442**. The rear ramped actuator **442** and the front ramped actuator **440** may engage the ramped upper sidewalls **412** and the ramped lower sidewalls **414** in the first and second portions **408** and **410** forcing the first and second portions **408** and **410** to expand from a collapsed position. A counter rotation of the central screw **440** may cause the front ramped actuator **440** and the rear ramped actuator **442** to separate causing the first and second portions **408** and **410** to collapse from the expanded state.

(148) Assembly of the barrel **406** of the medical device **400** will now be described according to an example implementation with reference to FIGS. **82-85**. As illustrated by FIG. **82**, the barrel **406** may comprise a first portion **408** and a second portion **410**. The first plate **402** may be defined by upper portion **420** and lower portion **422**. Upper portion **420** may extend from first portion **408** of the barrel **406**, and lower portion **422** may extend in an opposite direction from second portion **410** of the barrel **406**. As further illustrated by FIG. **83**, the actuator assembly **438** may comprise a central screw **440**, a front ramped actuator **442**, and a rear ramped actuator **444**. In FIG. **83**, the rear ramped actuator **444** may be slid onto the first portion **408** of the barrel **406**. As illustrated, the rear ramped actuator **444** may engage (e.g., through a dovetail connection) a rear end of the upper ramped sidewalls **412** of the first portion **408**. In FIG. **84**, the front ramped actuator **442** may then slide onto the second portion **410** of the barrel **406**. As illustrated, the ramped expansion portion **446** may engage (e.g., through a dovetail connection) a front end of the lower ramped sidewalls **414** of the second portion **410**. In FIG. **85**, the first portion **408** and second portion **410** of the barrel **406** have been placed together in a contracted position with a front end of the upper ramped sidewalls **412** engaging the ramped expansion portion **446** and a rear end of the lower ramped sidewalls **414** engaging the rear ramped actuator **444**.

(149) Referring to FIGS. **86-90**, the pivoting spike assembly **424** of the medical device **400** will now be described according to an example implementation. FIGS. **86-90** illustrate assembly of the pivoting spike assembly **424** in the first plate **402**. FIGS. **88-90** also show side, cross-sectional views and top views of the spike assembly **424**. As illustrated, the first plate **402** may include a pivoting spike assembly **424** on both the upper portion **420** and the lower portion **422**. The pivoting spike assemblies **424** may each be received within an opening **426** (best seen on FIG. **86**) in both the upper portion **420** and the lower portion **422**. Each pivoting spike assembly **424** may include multiple projections (e.g., spikes **428**). Each spike assembly **424** may comprise a base **429** with a hole **432** (best seen on FIG. **86**) for attachment to either the upper portion **420** or the lower portion **422** of the first plate **402**. The spike assembly **424** may further comprise leaf spring feature **431**.

(150) Assembly of the pivoting spike assembly **424** in the first plate **402** will now be described in accordance with an example embodiment. The following description describes insertion of the pivoting spike assembly **424** into the upper portion **420** of the first plate, but it should be understood that a pivoting spike assembly **424** may also be inserted into the lower portion **422** in a similar manner. As illustrated by FIGS. **86** and **87**, a fastener **433** may be inserted into the hole **432** to hold the spike assembly **424** to the first plate **402**. Once the pivoting spike assembly **424** is placed within the upper portion **420** and lower portion **422** of the first plate **402**, the hole **432** of the assembly **424** may be aligned with a hole **434** of the first plate **402**. As illustrated in FIG. **88** the fastener **433** may first be inserted in the hole **434** of the upper portion **420** and lower portion **422** of the first plate **402**. The fastener **433** may be configured in a cam-type configuration that includes a lobe **435**. Inserting the fastener **433** sideways allows the lobe **435** to pass through the hole **434** of the first plate **402**. The lobe **435** may come to rest on the pivoting spike assembly **424**. Rotating the fastener **433** ninety degrees, as seen in FIG. **89**, may force the lobe **435** into the leaf spring feature **431** causing it to expand. While the leaf spring feature **431** is expanded the fastener **433** may be advanced further into the hole **432** in the assembly **424**. Further advancement of the fastener **433**, as seen in FIG. **90**, may push the lobe **435** of the fastener **433** below the leaf spring feature **431** into a groove **436** found in the hole **432** of the pivoting spike assembly **424**. Once the lobe **435** is in the groove **436** of the pivoting spike assembly **424**, the leaf spring feature **431** may snap back into place against the fastener **433**. This may secure the fastener **433** to the pivoting spike assembly **424** and prevent detachment of the assembly **424** from the upper portion **420** or lower portion **422** of the first plate **402**. An audible sound of the leaf spring feature **431** snapping back into place may be used in some embodiments to alert the assembler that the medical device **400** has been properly put together.

(151) FIGS. **91** and **92** illustrate the pivoting spike assembly **424** secured in the first plate **402** in more detail in accordance with example embodiments. As illustrated in FIGS. **91** and **92**, the pivoting spike assembly **424** may be free to move around the fastener **433**. The fastener **433** and the inner hole **432** may have matching serrations **436** and **437** respectively. The leaf spring feature **431** may prevent serrations **436** and **437** from meshing, allowing the pivoting spike assembly **424** to be mobile. When compression force is applied to the spike assembly **424**, the leaf spring feature **431** may collapse upon the fastener **433**, allowing the serrations **436** and **437** to mesh and restrict movement as seen in FIG. **92**. When the implant is loosened or removed, the leaf spring feature **431** pushes against fastener **433**, effectively remobilizing the pivoting spike assembly **424** as seen in FIG. **91**.

(152) While the term “spikes” may be used for the projections in the pivoting spike assembly **424**, other types of projections may be used that may have a more tapered point or rounded point or other type of ending to the projection. The spikes **428** may be used to attach firmly and bite into the spinous processes above and below an interspinous space. The spikes **428** may be pyramid shaped with a base portion secured or integrally formed on the pivoting spike assembly **424**. The sides of the spikes **428** may extend from the base to form a point in the shape of a pyramid. In other example implementations, the spikes **428** may be formed into other shapes that rise to a point to enable the spike to engage the spinous process. As discussed above, the end of the spikes **428** may include tips other than a point such as, for example, rounded tip, a square tip or other-shaped tip. The example illustration of the medical device **400** includes three (3) spikes **428** on each pivoting spike assembly **424** of the first plate **402**. In other example implementations, fewer or more spikes **428** may be included. The first plate **402** and the spikes **428** may be made of titanium. In other implementations, the first plate **402** and the spikes **428** may be made of other biocompatible materials.

(153) Referring to FIGS. **93** and **94**, the second or locking plate **404** will now be described in more detail with respect to one example implementation. FIGS. **93** and **94** are side and front views of the locking plate **404** in accordance with example embodiments. The locking plate **404** may be inserted

onto the central screw **440** after the barrel **406** has been expanded to lock the barrel **406** in position, as best seen in FIG. **78**. As illustrated by FIGS. **93** and **94**, the locking plate **404** may comprise an upper portion **450** and a lower portion **452**. A central portion **454** may connect the upper portion **450** to the lower portion **452**. The upper portion **450**, lower portion **452**, and central portion **454** may be integrally formed as a single plate component in some embodiments. The central portion **454** includes an opening (e.g., a central opening) to receive trunion assembly **456**. The locking plate **404** may rotate about the trunion assembly **456** and can be locked at various angles at any position within its range of motion. In some embodiments, the trunion assembly **456** may be configured so that the locking plate **404** rotates about its center. The locking plate **404** may include spikes **428** on both the upper portion **450** and the lower portion **452**.

(154) With additional reference to FIGS. **95** and **96**, the trunion assembly **456** will be described in more detail with respect to one example implementation. As illustrated, the trunion assembly **456** may comprise a housing **468**. The housing **468** may have laterally extending projections **470** for rotatably coupling the trunion assembly **456** to the central portion **454** of the locking plate **404** while allowing the locking plate **404** to rotate with respect to the trunion assembly **468**. As illustrated, there may be a pair of projections **470** that extend from opposite sides of the housing **468** and are each received in corresponding openings **472** (best seen on FIG. **93**) in the central portion **454**. The housing **468** may further have a through bore **474** for receiving the central screw **440**. The housing **468** may further comprise a pair of chambers **476** on either side of the through bore **474**. The housing **468** may further include a ratchet pawl **478** in each chamber **476**.

Embodiments of the ratchet pawls **478** may be spring loaded so that the ratchet pawls **478** may maintain contact with the central screw **440** while the locking plate **404** rotates about the trunion assembly **468**. The ratchet pawls **478** may be assembled from the side of the housing **468**. The ratchet pawls **478** may each have spring cuts to allow the ratchet pawls **478** to compress further into the chambers **476**. The spring cuts may be the height of an electric discharge machining wire to create a small gap within each leaf of the ratchet pawls **478** being self-limiting as it collapses upon itself. Insertion of the central screw **440** into the through bore **474** (e.g., from right to left of FIG. **78**) should cause the teeth (or threading) of the central screw **440** to engage the ratchet pawls **478** causing the ratchet pawls **478** to recess into the chambers **476**. The angling of the teeth on the ratchet pawls **478** should resist backwards motion of the central screw **440** after insertion into the through bore **474**. In this manner, the ratchet pawls **478** may be operable to secure the trunion assembly **468** and thus the locking plate **404** onto the central screw **440**.

(155) FIG. **97** illustrates a tube **480** that can be used to release the ratchet pawls **478** in accordance to one example implantation. The tube **480** may be sized to fit over the central screw **440**. The tube **480** may be advanced over the central screw **440** and into the back end of the through bore **474** until the leading end or nose **482** of the tube **480** engages the ratchet pawls **478**. Pressure from the tube **480** combined with large chamfers on the ratchet pawls should cause the ratchet pawls **478** to compress. When fully inserted, the tube **480** includes one or more teeth **484** configured to snap into the ratchet pawls **478** allowing complete release of the central screw **440**.

(156) As illustrated by FIG. **96**, the housing **468** may have an upper surface **486** and a lower surface **488**. In embodiments, the upper and lower surfaces **486** and **488** may each be curved. As illustrated, the upper and lower surfaces **486** and **488** may be sloped inward from the rear to the front of the housing **468**. In some embodiments, the upper and lower surfaces **486** and **488** may each comprise a projection **490**. The projection **490** may engage the locking plate **404** to limit its rotation about the trunion assembly **456**.

(157) Referring to FIGS. **98-100**, assembly of the locking plate **404** shown on FIGS. **93** and **94** will now be described according to an example implementation. As illustrated, the locking plate **404** may comprise an upper portion **450**, a lower portion **452**, and a central portion **454** coupling the upper portion **450** and the lower portion **452**. The trunion assembly **456** may comprise a housing **468** and a pair of ratchet pawls **478**. The housing **468** may comprise a pair of windows **492** for

receiving the ratchet pawls **478** into chambers **476** (FIG. **95**). The ratchet pawls **478** may be inserted into the housing **468** from the side via windows **492**, as shown on FIG. **99**. The trunion assembly **456** comprising the housing **468** having the ratchet pawls **478** disposed therein may then be inserted into the opening **479** in the central portion **454** of the locking plate **404**, as best seen in FIG. **100**.

(158) As previously mentioned, the locking plate **404** may be free to rotate about the trunion assembly **456** even where the trunion assembly **456** is in engagement with central screw **440**. FIG. **101** is a view of the medical device **400** showing rotation of the locking plate **404** according to one example implementation. The locking plate **404** may also include a pivoting spike assembly **424** that may be assembled and function in a manner similar to that described above for the first plate **402**. As illustrated, the pivoting spike assemblies **424** of the first plate **402** and the locking plate **404** may also be free to articulate with respect to the locking plate **404**. Rotation of the locking plate **404** and/or articulation of the pivoting spike assemblies **424** can provide an adaptable medical device **400** that can accommodate variances in spinous process geometry, for example, with the goal of anterior and secure placement.

(159) An embodiment for using the medical device **400** will now be described in accordance with one example implementation. For example, a method may comprise inserting the barrel **406** of the medical device **400** into an interspinous space. The method may further comprise expanding the barrel **406** from a collapsed form having a first height to an expanded form having a second height, where the second height is greater than the first height. As discussed above, the central screw **440** may be rotated to expand the barrel **406** from a collapsed form to an expanded form in the interspinous space. The process may further include inserting the locking plate **404** onto the central screw **440** and moving the locking plate **404** towards the first plate such that the locking plate **404** and the first plate **402** engage a spinous process. The locking plate **404** may be free to rotate about its center (e.g., the trunion assembly **456**) to accommodate spinous process geometry. In addition, the pivoting spike assembly **424**, the first plate **402**, and the locking plate **404**, respectively, may also be free to articulate for accommodation of spinous process geometry. The pivoting spike assembly **424** may be locked into place during compression into the spinous process.

(160) The various components of the medical device **10**, medical device **100**, medical device **300**, and medical device **400**, described herein can be formed with any biocompatible material used for such a medical device. For example, each of the various components can be formed with one or more biocompatible plastics and/or one or more biocompatible metals such as, for example, titanium and stainless steel.

(161) While certain features of the described implementations have been illustrated as described herein, many modifications, substitutions, changes and equivalents will now occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the scope of the embodiments. It should be understood that they have been presented by way of example only, not limitation, and various changes in form and details may be made. Any portion of the apparatus and/or methods described herein may be combined in any combination, except mutually exclusive combinations. The embodiments described herein can include various combinations and/or sub-combinations of the functions, components and/or features of the different embodiments described.

Claims

1. A method for implanting a medical device in a patient, comprising: inserting a medical device including: a barrel having an upper portion and a lower portion; an actuator assembly disposed in the barrel, the actuator assembly including a front ramped actuator in engagement with the barrel, a rear ramped actuator in engagement with the barrel, and a central screw that extends from the rear ramped actuator through the front ramped actuator; and a first plate having a first portion that

extends from the upper portion, a second portion that extends from the lower portion, and a first pivoting spike assembly disposed in the first plate and adapted to pivot relative to the first plate, the first pivoting spike assembly including one or more spikes extending from a first side of the first plate; rotating the central screw disposed in the barrel to cause the barrel to transition from a collapsed form having a first height to an expanded form having a second height; inserting onto the central screw a second plate having multiple spikes extending from a first side of the second plate; and pivotally attaching the first pivoting spike assembly to the first plate with a fastener, wherein the first pivoting spike assembly includes a spring disposed to face the fastener such that when the spring is in an uncompressed state, the first pivoting spike assembly is free to move about the fastener.

2. The method of claim 1, wherein when a compression force applied to the first pivoting spike assembly, the spring collapses upon the fastener to immobilize the first pivoting spike assembly relative to the first plate.

3. The method of claim 2, wherein the first pivoting spike assembly includes serrations and the fastener includes corresponding serrations that mesh with the serrations of the first pivoting spike assembly to immobilize the first pivoting spike assembly relative to the first plate when the compression force is applied to the first pivoting spike assembly.

4. The method of claim 1, further comprising movably attaching the first pivoting spike assembly to the first plate with a fastener, wherein: the first pivoting spike assembly includes a spring disposed to face the fastener such that when the spring is in an uncompressed state, the first pivoting spike assembly is free to move about the fastener, and when a compression force is applied to the first pivoting spike assembly, the spring collapse upon the fastener to immobilize the first pivoting spike assembly relative to the first plate.

5. The method of claim 4, wherein the first pivoting spike assembly includes serrations and the fastener includes corresponding serrations that mesh with the serrations of the first pivoting spike assembly to immobilize the first pivoting spike assembly relative to the first plate when the compression force is applied to the first pivoting spike assembly.

6. The method of claim 1, wherein the upper portion of the barrel comprises ramped upper sidewalls in engagement with the rear ramped actuator and the front ramped actuator, and wherein the lower portion of the barrel comprises ramped lower sidewalls in engagement with the rear ramped actuator and the front ramped actuator.

7. The method of claim 6, wherein the ramped upper sidewalls and the ramped lower sidewalls define a central opening configured to receive graft packing material.

8. The method of claim 1, wherein the first portion of the first plate is integrally formed with the upper portion of the barrel, and wherein the second portion of the first plate is integrally formed with the lower portion of the barrel.

9. The method of claim 1, wherein the second plate is a locking plate.

10. A method for implanting a medical device in a patient, comprising: inserting a medical device including: a barrel, the barrel having an upper portion and a lower portion; an actuator assembly disposed in the barrel; a first plate, formed integrally with the barrel, including a first portion that extends from the upper portion, a second portion that extends from the lower portion, and a first pivoting spike assembly disposed in the first plate and adapted to pivot relative to the first plate, the first pivoting spike assembly including one or more spikes extending from a first side of the first plate; and a locking plate having multiple spikes extending from a first side of the locking plate, the locking plate configured to be received on a central screw disposed in the actuator assembly; rotating the central screw disposed in the barrel to cause the barrel to transition from a collapsed form having a first height to an expanded form having a second height; inserting onto the central screw the locking plate; and pivotally attaching the first pivoting spike assembly to the first plate with a fastener, wherein the first pivoting spike assembly includes a spring disposed to face the fastener such that when the spring is in an uncompressed state, the first pivoting spike assembly is

free to move about the fastener.

11. The method of claim 10, wherein when a compression force applied to the first pivoting spike assembly, the spring collapses upon the fastener to immobilize the first pivoting spike assembly relative to the first plate.

12. The method of claim 11, wherein the first pivoting spike assembly includes serrations and the fastener includes corresponding serrations that mesh with the serrations of the first pivoting spike assembly to immobilize the first pivoting spike assembly relative to the first plate when the compression force is applied to the first pivoting spike assembly.

13. The method of claim 10, further comprising movably attaching the first pivoting spike assembly to the first plate with a fastener, wherein: the first pivoting spike assembly includes a spring disposed to face the fastener such that when the spring is in an uncompressed state, the first pivoting spike assembly is free to move about the fastener, and when a compression force is applied to the first pivoting spike assembly, the spring collapse upon the fastener to immobilize the first pivoting spike assembly relative to the first plate.

14. The method of claim 13, wherein the first pivoting spike assembly includes serrations and the fastener includes corresponding serrations that mesh with the serrations of the first pivoting spike assembly to immobilize the first pivoting spike assembly relative to the first plate when the compression force is applied to the first pivoting spike assembly.

15. The method of claim 10, wherein the actuator assembly comprises a front ramped actuator and a rear ramped actuator, and wherein the upper portion of the barrel comprises ramped upper sidewalls in engagement with the rear ramped actuator and the front ramped actuator, and wherein the lower portion of the barrel comprises ramped lower sidewalls in engagement with the rear ramped actuator and the front ramped actuator.

16. The method of claim 15, wherein the ramped upper sidewalls and the ramped lower sidewalls define a central opening configured to receive graft packing material.
